

Daeseon Yoo

Software Engineer | **Authorized to work in Canada** | +1 (437) 218-7432 | **Toronto, ON** | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Blog](#)

PROFESSIONAL SUMMARY

Backend Engineer with **6** years of experience designing and operating data-heavy enterprise systems in global manufacturing environments. Specialized in ensuring transaction integrity, concurrency control, and system reliability using Java, C#, and SQL. Currently applying production-grade fault tolerance—such as async workflows, retry/backoff, and state-machine execution—to build robust infrastructure for AI-enabled applications.

TECHNICAL SKILLS

Languages: Python, Java, Go, TypeScript, C#, SQL

Backend: Spring Boot, FastAPI, .NET, JPA, MyBatis, REST APIs, gRPC

Frontend: React, Next.js, Tailwind CSS

Data, Search & Messaging: PostgreSQL, Oracle, Redis, Kafka, RabbitMQ, SQL performance tuning

Infra & Testing: Docker, AWS, GitHub Actions, Terraform, k6, Testcontainers, Playwright, Vitest

AI: LLM APIs, RAG, embeddings, vector search, function calling, MCP

WORK EXPERIENCE

Software Engineer | SK AX

Sep. 2021 - May 2026

- Redesigned transaction boundaries between manufacturing operations and financial systems using a staging-table and batch-worker pattern, cutting processing time by **60%** and reducing reprocessing from **20 monthly cases to near zero**.
- Eliminated cross-server duplicate execution with a distributed lock (ShedLock/job_lock table), and decoupled the scheduler from execution via @Async so both month-end plant-close jobs run in parallel.
- Resolved duplicate ID generation across multi-instance manufacturing execution services by implementing Oracle row-level locks (FOR UPDATE), preventing data anomalies and manufacturing downtime.
- Consolidated **46 middleware endpoints** into a single unified API using a reflection-based RPC dispatcher, deployed across **8 sites** — freeing **1 FTE** of middleware development by eliminating redeployments.
- Resolved HTTP 413 failures in mobile scan flows without infrastructure changes by replacing full payload submissions with PK-only requests and server-side fresh-state retrieval, raising batch size from **30 to 80** while keeping inventory accurate.
- Implemented a polymorphic architecture to dynamically dispatch site-specific logic without hardcoded conditionals, scaling execution across **8 plants** with no added central headcount.
- Refactored a **7.7K-line** monolithic CASE-based PL/SQL query into per-process CTEs, reducing query size by **57%**, isolating **5** sub-processes, and raising weekly cost-ledger ticket throughput from **4 to 7**.

Software Developer | Dure Info

Jun. 2020 - Sep. 2021

- Built and supported TCP socket server workflows for PC, PDA, and kiosk clients using .NET, enabling server-side stored procedure execution, DB credential isolation, and application-level message framing for partial TCP reads, which improved system reliability and reduced client-side connection errors.

SELECTED PROJECTS

Talkak | AI-agent workspace and operating-memory system | Rust · Tauri · React · tmux

[Live Demo](#) | [Deep Dive](#)

- Developed a Tauri/Rust desktop process manager for AI/CLI workflows, using tmux sessions to preserve long-running subprocess state across React component remounts and app navigation.
- Built an append-only local JSONL event/graph store for work tracking, storing work items, actions, decisions, evidence, approvals, and verification receipts with project-scoped retrieval.

Mimi - YouTube-clip English shadowing trainer | Spring Boot · Next.js · PostgreSQL · AWS

[Live Demo](#) | [Deep Dive](#)

- Built an event-driven async LLM pipeline using Spring AFTER_COMMIT and bounded @Async so provider calls run outside DB transactions, eliminating connection-pool exhaustion with PENDING to READY/FAILED states.
- Implemented multi-provider fallback with retry/backoff for 429/5xx failures, isolating provider throttling from workflows.

DocVault - Windows endpoint audit platform | Go · PostgreSQL · Windows Agent

[Live Demo](#) | [Deep Dive](#)

- In active use by a 40-person team**; built Windows agent onboarding with single-use install tokens, server-side token provisioning, automatic host-user mapping, heartbeat checks, and telemetry self-tests.
- Implemented tamper-evident endpoint audit workflows using osquery and custom endpoint telemetry, PostgreSQL hash-chained logs, RBAC file-vault access, alert rules, and admin views for offline, unassigned, and unverified PCs.

EDUCATION

M.S. & B.S. Industrial Engineering - Kumoh National Institute of Technology, South Korea

Mar. 2012 - Feb. 2019